

100% Excellent!

PHY201 Exam on chapter(s): 11, 12

These questions assess your understanding of the material from Chapters 11, 12 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. The aorta pumps blood from the heart to the rest of the body. Calculate the average speed of the blood in Daniel's aorta if the flow rate of his blood is 12.1 L/min. Daniel's aorta has a radius of 11 mm.

$$\frac{12.1 \text{ L}}{1 \text{ min}} \times \frac{1 \text{ m}^3}{1000 \text{ L}} \times \frac{1 \text{ min}}{60 \text{ s}} = 2.017 \times 10^{-4} \frac{\text{m}^3}{\text{s}} = Q$$

ANSWER: $0.531 \frac{\text{m}}{\text{s}}$

$$Q = A\bar{v} = \pi r^2 \bar{v} \rightarrow \bar{v} = \frac{Q}{\pi r^2}$$

$$\bar{v} = \frac{2.017 \times 10^{-4} \frac{\text{m}^3}{\text{s}}}{\pi (0.011 \text{ m})^2} = \boxed{0.531 \frac{\text{m}}{\text{s}}}$$

2. A reservoir has a surface area of 62 km² and an average depth of 27 m. What mass of water is held behind the dam?

$$\frac{62 \text{ km}^2}{1^2 \text{ km}^2} \times \frac{1000^2 \text{ m}^2}{1^2 \text{ km}^2} = 6.2 \times 10^7 \text{ m}^2 = A$$

ANSWER: $1.674 \times 10^{12} \text{ kg}$

$$\rho_w = \frac{m}{V} = \frac{m}{Ah} \rightarrow m = Ah\rho$$

$$m = (6.2 \times 10^7 \text{ m}^2)(27 \text{ m})(1000 \frac{\text{kg}}{\text{m}^3}) = \boxed{1.674 \times 10^{12} \text{ kg}}$$

3. A particular dam is 690-m wide and contains water that is 174-m deep. What is the average pressure on the dam due to the water?

ANSWER: 852600 Pa

$$P_2 = P_1 + \rho_w gh = P + \cancel{P_{atm}} = \cancel{P_{atm}} + \rho_w gh$$

$$P = (1000 \frac{\text{kg}}{\text{m}^3})(9.8 \frac{\text{m}}{\text{s}^2})(\frac{174}{2} \text{ m}) = \boxed{852600 \text{ Pa}}$$

+3

4. Kevin and Leah are on a slowly sinking ship, but they have hope: Kevin managed to find an oaken door that is 223-cm tall, 95-cm wide, and 7-cm deep. Perhaps they could both fit on the door if he throws it into the sea. What is the maximum mass the door will support? Leah recently took a physics class and remembers the density of oak is about 870 kg/m^3 and the density of seawater is 1025 kg/m^3 .

ANSWER: 22.99 kg

$$f_{\text{sub}} = \frac{\rho_{\text{oak}}}{\rho_{\text{H}}} = \frac{\rho_{\text{o}} + \rho_{\text{o}}}{\rho_{\text{s}}} = \frac{\frac{m_{\text{o}}}{V_{\text{o}}} + \rho_{\text{o}}}{\rho_{\text{s}}} \rightarrow \rho_{\text{s}} f_{\text{sub}} = \frac{m_{\text{o}}}{V_{\text{o}}} + \rho_{\text{o}}$$

$$m_{\text{o}} = V_{\text{o}}(\rho_{\text{s}} f_{\text{sub}} - \rho_{\text{o}})$$

$$m_{\text{o}} = (2.23 \text{ m})(0.95 \text{ m})(0.07 \text{ m}) \left((1025 \frac{\text{kg}}{\text{m}^3})(1) - 870 \frac{\text{kg}}{\text{m}^3} \right) = \boxed{22.99 \cdot \text{kg}}$$

5. Emma's doctor performs a special test and tells her that a 3.0-N force can rupture her eardrum. If her eardrum has an area of 1.09 cm^2 , at what depth in freshwater would her eardrum rupture, assuming the gauge pressure in the middle ear is zero and she does nothing to alleviate the pressure?

ANSWER: 2.81 m

$$\frac{1.09 \text{ cm}^2}{100^2 \frac{\text{cm}^2}{\text{m}^2}} = 1.09 \times 10^{-4} \text{ m}^2 = A$$

$$p = \frac{F}{A} = \rho_{\text{w}} g h \rightarrow h = \frac{F}{A \rho_{\text{w}} g}$$

$$h = \frac{3 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}}{(1.09 \times 10^{-4} \text{ m}^2)(1000 \frac{\text{kg}}{\text{m}^3})(9.8 \frac{\text{m}}{\text{s}^2})} = \boxed{2.81 \text{ m}}$$

6. Efe attaches a nozzle of diameter 6.0 mm to a garden hose of diameter 2.4 cm. If the hose fills up a 2-L bucket in 13.0 seconds, calculate the pressure in the hose assuming level, frictionless flow.

ANSWER: 116070.47 Pa

$$\frac{2 \text{ L}}{13 \text{ s}} \frac{1 \text{ m}^3}{1000 \text{ L}} = 1.538 \times 10^{-4} \frac{\text{m}^3}{\text{s}} = Q$$

$$P_1 + \frac{1}{2} \rho_{\text{w}} v_1^2 + \rho_{\text{w}} g h_1 = P_2 + \frac{1}{2} \rho_{\text{w}} v_2^2 + \rho_{\text{w}} g h_2$$

$$Q = A \bar{v} \rightarrow \bar{v} = \frac{Q}{A} = \frac{Q}{\pi r^2} = \frac{Q}{\pi (\frac{d}{2})^2} = \frac{4Q}{\pi d^2}$$

$$P_1 = P_2 + \frac{1}{2} \rho_{\text{w}} v_2^2 - \frac{1}{2} \rho_{\text{w}} v_1^2 = P_2 + \frac{1}{2} \rho_{\text{w}} \left(\left(\frac{4Q}{\pi d_2^2} \right)^2 - \left(\frac{4Q}{\pi d_1^2} \right)^2 \right) = P_2 + \frac{16Q^2 \rho_{\text{w}}}{2\pi^2} \left(\frac{1}{d_2^4} - \frac{1}{d_1^4} \right)$$

$$P_1 = 101325 \frac{\text{kg}}{\text{m} \cdot \text{s}^2} + \frac{16(1.538 \times 10^{-4} \frac{\text{m}^3}{\text{s}})^2 (1000 \frac{\text{kg}}{\text{m}^3})}{2\pi^2} \left(\frac{1}{(0.006 \text{ m})^4} - \frac{1}{(0.024 \text{ m})^4} \right)$$

$$P_1 = \boxed{116070.47 \text{ Pa}}$$

7. Hannah is an astronaut working outside the International Space Station (ISS) where the atmospheric pressure is essentially zero. The pressure gauge on the astronaut's air tank reads 6.8×10^6 Pa. What force does the air inside the tank exert on the flat end of the cylindrical tank, a disk that is 0.19 m in diameter?

ANSWER: 192799.54 N ✓

$$P = \frac{F}{A} = \frac{F}{\pi r^2} = \frac{F}{\pi \left(\frac{d}{2}\right)^2} = \frac{F}{\frac{\pi d^2}{4}} = \frac{4F}{\pi d^2}$$

$$F = \frac{\pi d^2 P}{4}$$

$$F = \frac{\pi (0.19 \text{ m})^2 (6.8 \times 10^6 \frac{\text{N}}{\text{m}^2})}{4} = \boxed{192799.54 \text{ N}}$$

8. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Isabella's aorta is 16.0 L/min and Isabella's aorta has a radius of 14 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 9.6 μm . The speed of blood in the capillaries is about 0.27 mm/s. Calculate the number of capillaries in Isabella's blood circulatory system.

ANSWER: 1.364×10^{10} capillaries ✓

$$\frac{16 \text{ L}}{1 \text{ min}} \left| \frac{1 \text{ m}^3}{1000 \text{ L}} \right| \frac{1 \text{ min}}{60 \text{ s}} = 2.667 \times 10^{-4} \frac{\text{m}^3}{\text{s}} = Q$$

$$9.6 \mu\text{m} = 9.6 \times 10^{-6} \text{ m} = d_2$$

$$n_1 A_1 \bar{v}_1 = n_2 A_2 \bar{v}_2 = n_1 Q = n_2 \pi r_2^2 \bar{v}_2 = n_2 \pi \left(\frac{d_2}{2}\right)^2 \bar{v}_2$$

$$n_2 = \frac{n_1 Q}{\pi \left(\frac{d_2}{2}\right)^2 \bar{v}_2}$$

$$n_2 = \frac{(1)(2.667 \times 10^{-4} \frac{\text{m}^3}{\text{s}})}{\pi \left(\frac{9.6 \times 10^{-6} \text{ m}}{2}\right)^2 (0.00027 \frac{\text{m}}{\text{s}})} = \boxed{1.364 \times 10^{10} \text{ capillaries}}$$

9. Calculate the maximum buoyant force that water could exert on a ship with a mass of 1.3×10^8 kg if the ship displaces $6.0 \times 10^6 \text{ m}^3$ of water.

ANSWER: $5.88 \times 10^{10} \text{ N}$ ✓

$$F_b = F_{g_w} = m_w g$$

$$\rho_w = \frac{m_w}{V_s} \rightarrow m_w = V_s \rho_w$$

$$F_b = V_s \rho_w g$$

$$F_b = (6 \times 10^6 \text{ m}^3) (1000 \frac{\text{kg}}{\text{m}^3}) (9.8 \frac{\text{m}}{\text{s}^2}) = \boxed{5.88 \times 10^{10} \text{ N}}$$

10. Victoria creates a soap bubble and measures its diameter to be $245 \mu\text{m}$. Calculate the absolute pressure inside of the bubble, given that the surface tension of typical soapy water is 0.0370 N/m . Report your answer in mmHg, recalling that 133.32 Pa is 1 mmHg.

$$245 \mu\text{m} = 245 \times 10^{-6} \text{ m} = d$$

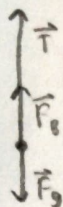
ANSWER: 769.08 mmHg

$$P = P_g + P_{\text{atm}} = \frac{4\gamma}{r} + P_{\text{atm}} = \frac{4\gamma}{\frac{d}{2}} + P_{\text{atm}}$$

$$P = \frac{4(0.037 \frac{\text{N}}{\text{m}})}{245 \times 10^{-6} \text{ m}} + 101325 \frac{\text{N}}{\text{m}^2} = 102533.163 \text{ Pa}$$

$$\frac{102533.163 \text{ Pa}}{133.32 \text{ Pa}} = 769.08 \text{ mmHg}$$

11. While Gabriel is fishing, he begins to reel in a reel big fish. His "FishFinder 3000" tells him that this particular fish has a volume of 8800 cm^3 . After Gabriel catches the fish and weighs it, he finds it has a mass of 9.69 kg . What was the minimum tension force in the fishing line if the fish was pulled vertically upwards through freshwater, which has a density of 1000 kg/m^3 ? You may ignore the effects of the drag force.



$$\frac{8800 \text{ cm}^3}{1000 \text{ cm}^3} = 0.0088 \text{ m}^3 = V_f$$

ANSWER: 8.722 N

$$\Sigma F_y = T + F_b - F_g = m a_y \rightarrow T = F_g - F_b$$

$$F_b = m_w g$$

$$\rho_w = \frac{m_w}{V_f} \rightarrow m_w = V_f \rho_w$$

$$F_b = V_f \rho_w g$$

$$T = m_f g - V_f \rho_w g = g(m_f - V_f \rho_w)$$

$$T = (9.8 \frac{\text{m}}{\text{s}^2})(9.69 \text{ kg} - (0.0088 \text{ m}^3)(1000 \frac{\text{kg}}{\text{m}^3})) = 8.722 \text{ N}$$

12. Jennifer attaches a nozzle of diameter 6.8 mm to a garden hose of diameter 1.6 cm . If the hose fills up a 1-L bucket in 13.3 seconds, what is the speed of the water that leaves the nozzle?

$$\frac{1 \text{ L}}{13.3 \text{ s}} = \frac{1 \text{ m}^3}{1000 \text{ L}} = 7.519 \times 10^{-5} \frac{\text{m}^3}{\text{s}} = Q$$

ANSWER: $2.07 \frac{\text{m}}{\text{s}}$

$$Q = A \bar{v} = \pi r^2 \bar{v} = \pi \left(\frac{d}{2}\right)^2 \bar{v} = \frac{\pi d^2 \bar{v}}{4} \rightarrow \bar{v} = \frac{4Q}{\pi d^2}$$

$$\bar{v} = \frac{4(7.519 \times 10^{-5} \frac{\text{m}^3}{\text{s}})}{\pi(0.0068 \text{ m})^2} = 2.07 \frac{\text{m}}{\text{s}}$$